

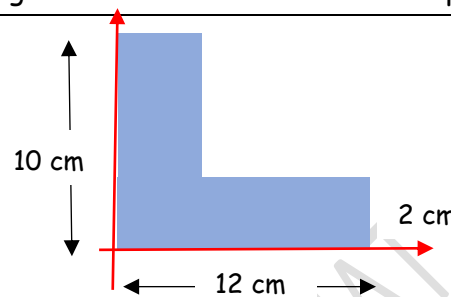
Moment of inertia of composite shapes

By knowing the moment of inertia of the basic shapes like rectangle, triangle, circle etc., we can find out the moment of inertia of a composite shape by dividing it into the elemental basic shapes.

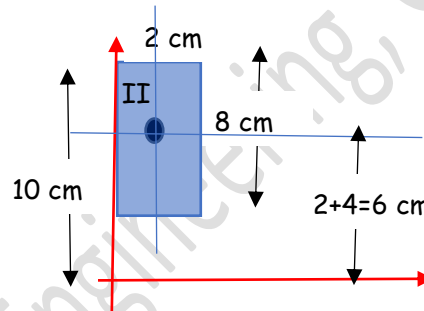
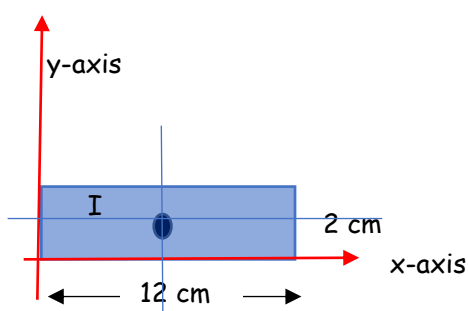
Example 1

Calculate the moment of inertia of the angle section shown with respect to the given x-y axes.

Let us divide the given shape into two rectangles as shown. For rectangle I, both the given x-y axes pass through its edges. Hence, we can use the formula we derived earlier for I_{xx} and I_{yy} .



But, for rectangle II, only the y-axis passes through its edge. To find I_{xx} , we need to use the parallel axis theorem.



Moment of inertia of rectangle I with respect to the given x-axis, $I_{1xx} = \frac{bh^3}{3} = \frac{12 \times 2^3}{3} = 32 \text{ cm}^4$

Moment of inertia of rectangle I with respect to the given y-axis, $I_{1yy} = \frac{b^3h}{3} = \frac{12^3 \times 2}{3} = 1152 \text{ cm}^4$

Since the x-axis does not pass either through the bottom edge or through the centroid of II (we know the expressions for them), we need to apply the parallel axis theorem.

M.I. of rectangle II with respect to the given x-axis = M.I. of rectangle II w.r.t. its parallel centroidal axis + (area of rectangle $\times$ square of the distance between the two axes)

$$I_{2xx} = \overline{I_{2xx}} + A_2 (6^2) = \frac{bh^3}{12} + (2 \times 8)(6^2) = \frac{2 \times 8^3}{12} + (16 \times 36) = 661.3 \text{ cm}^4$$

M.I. of rectangle II with respect to the given y-axis, $I_{2yy} = \frac{b^3h}{3} = \frac{2^3 \times 8}{3} = 21.3 \text{ cm}^4$

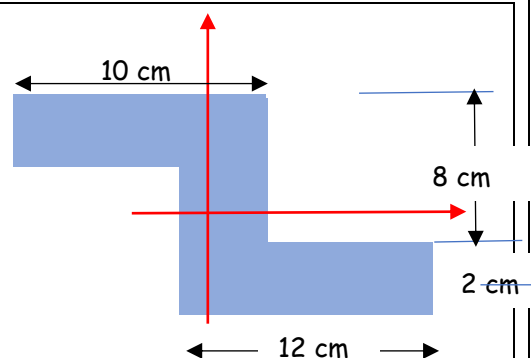
$$I_{xx} = I_{1xx} + I_{2xx} = 32 + 661.3 = 693.3 \text{ cm}^4$$

$$I_{yy} = I_{1yy} + I_{2yy} = 1152 + 21.3 = 1173.3 \text{ cm}^4$$

#Example 2

Calculate the moment of inertia of the area with respect to the x-y axes passing through its centroid. The width of the section is 2 cm throughout.

Let us locate the centroid of the given shape with respect to any co-ordinate system. Consider a co-ordinate system as shown passing through the left and bottom ends. You can fix any co-ordinate system of your choice to locate the centroid of the figure. Accordingly, you will get the values of the co-ordinates of the centroid.

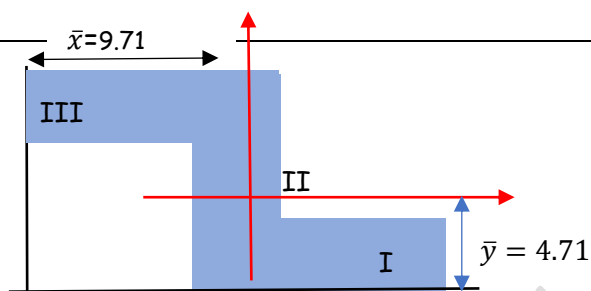
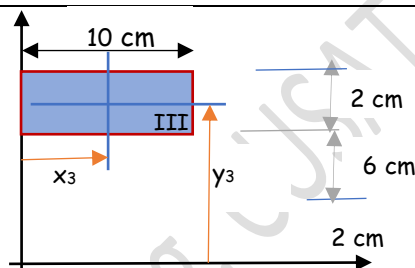
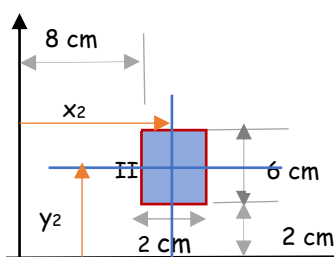
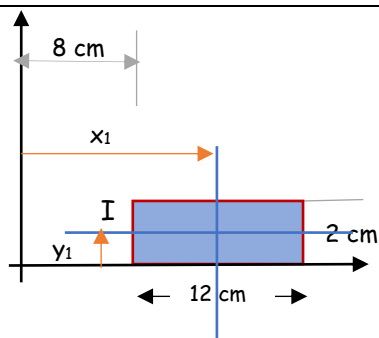
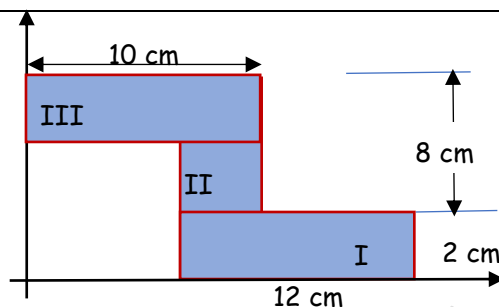


Divide the given shape in to three rectangles as shown

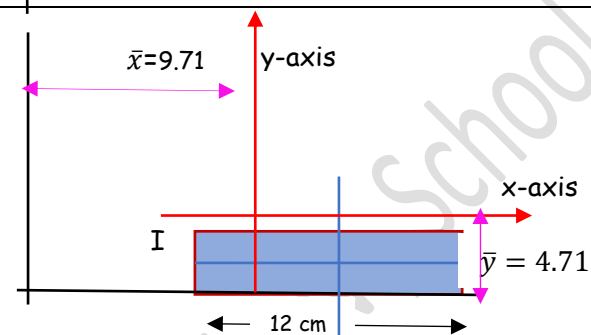
$A_1 = 12 \times 2 = 24 \text{ cm}^2$	$x_1 = 8 + 6 = 14 \text{ cm}$	$y_1 = 1 \text{ cm}$
$A_2 = 6 \times 2 = 12 \text{ cm}^2$	$x_2 = 8 + 1 = 9 \text{ cm}$	$y_2 = 2 + 3 = 5 \text{ cm}$
$A_3 = 10 \times 2 = 20 \text{ cm}^2$	$x_3 = 5 \text{ cm}$	$y_3 = 10 - 1 = 9 \text{ cm}$

$$\bar{x} = \frac{A_1 x_1 + A_2 x_2 + A_3 x_3}{A_1 + A_2 + A_3} = 9.71 \text{ cm}$$

$$\bar{y} = \frac{A_1 y_1 + A_2 y_2 + A_3 y_3}{A_1 + A_2 + A_3} = 4.71 \text{ cm}$$



The centroid of the given shape is at (9.71cm, 4.71cm) from the left and bottom edges. Now let us fix the co-ordinate system at the centroid, with respect to which the moment of inertia is to be found out.



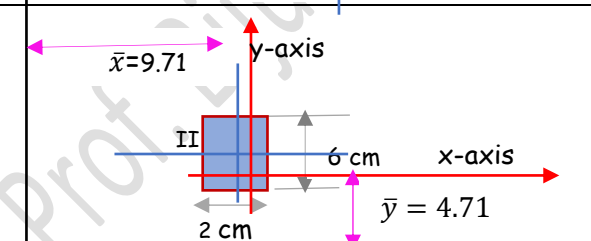
The given axes are marked in red. We need to apply the parallel axes theorem to find the MI of the rectangle 1 w.r.t the x-axis

$$I_{1xx} = \bar{I}_{1xx} + A_1 (\bar{y} - y_1)^2$$

$$= \frac{12 \times 2^3}{12} + (24)(4.71 - 1)^2 = 338.3 \text{ cm}^4$$

$$I_{1yy} = \bar{I}_{1yy} + A_1 (x_1 - \bar{x})^2$$

$$= \frac{12^3 \times 2}{12} + (24)(14 - 9.71)^2 = 729.69 \text{ cm}^4$$

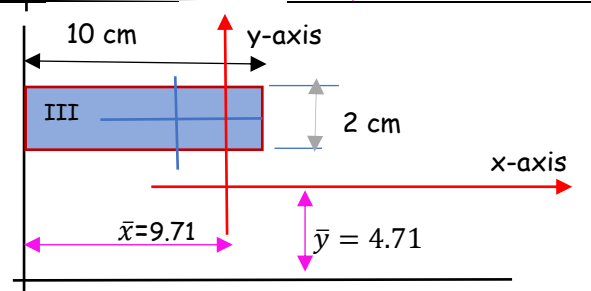


$$I_{2xx} = \bar{I}_{2xx} + A_2 (y_2 - \bar{y})^2$$

$$= \frac{2 \times 6^3}{12} + (12)(5 - 4.71)^2 = 37 \text{ cm}^4$$

$$I_{2yy} = \bar{I}_{2yy} + A_2 (x_2 - \bar{x})^2$$

$$= \frac{2^3 \times 6}{12} + (12)(9.71 - 9)^2 = 10.05 \text{ cm}^4$$



$$I_{3xx} = \bar{I}_{3xx} + A_3 (y_3 - \bar{y})^2$$

$$= \frac{10 \times 2^3}{12} + (20)(9 - 4.71)^2 = 374.75 \text{ cm}^4$$

$$I_{3yy} = \bar{I}_{3yy} + A_3 (\bar{x} - x_3)^2$$

$$= \frac{10^3 \times 2}{12} + (20)(9.71 - 5)^2 = 610.34 \text{ cm}^4$$

$$I_{xx} = I_{1xx} + I_{2xx} + I_{3xx} = 750 \text{ cm}^4$$

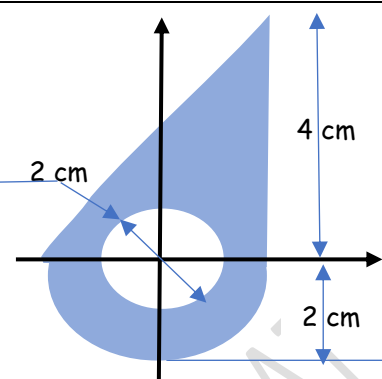
$$I_{yy} = I_{1yy} + I_{2yy} + I_{3yy} = 1350 \text{ cm}^4$$

Example 3

Calculate the moment of inertia of the shaded region with respect to the x-y axes shown.

We can divide the given figure into three elemental shapes. A right-angled triangle plus a semi-circle of radius 2 cm minus a circle of diameter 2 cm.

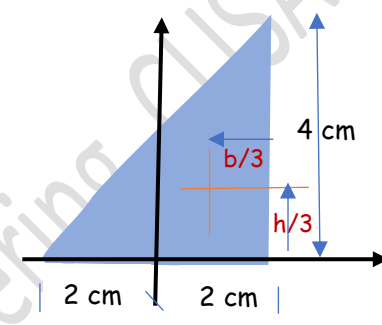
$$A_1 = \frac{1}{2} \times 4 \times 4 = 8 \text{ cm}^2, A_2 = \frac{1}{2} \pi \times 2^2 = 6.28 \text{ cm}^2, A_3 = \pi \times 1^2 = 3.14 \text{ cm}^2$$



We know the MI of a right triangle about the x-y axes passing through the right-angled edges and also when they pass through the centroid. Here the y-axis does not pass either through the vertical edge or through the centroid. Hence, we need to apply the parallel axis theorem to find I_{1yy}

$$I_{1xx} = \frac{bh^3}{12} = \frac{4 \times 4^3}{12} = 21.33 \text{ cm}^4$$

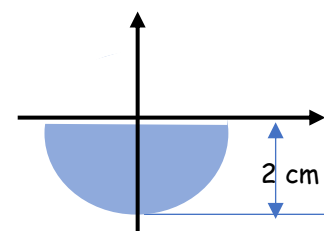
$$I_{1yy} = \overline{I_{1yy}} + A_1 \left(2 - \frac{b}{3}\right)^2 = \frac{4^3 \times 4}{36} + (8) \left(2 - \frac{4}{3}\right)^2 = 10.67 \text{ cm}^4$$



The moment of inertia of a semi-circle about the x-axis or y axis = 2 times the moment of inertia of a quarter circle about the same x-axis or y-axis.

$$I_{2xx} = 2 \times \frac{\pi r^4}{16} = \frac{\pi r^4}{8} = \frac{\pi \times 2^4}{8} = 6.28 \text{ cm}^4 \quad (\text{same as half of the MI of a circle about its diameter})$$

$$I_{2yy} = 2 \times \frac{\pi r^4}{16} = \frac{\pi r^4}{8} = \frac{\pi \times 2^4}{8} = 6.28 \text{ cm}^4$$



The moment of inertia of a circular area with respect to any of its diameter = $\frac{\pi r^4}{4}$

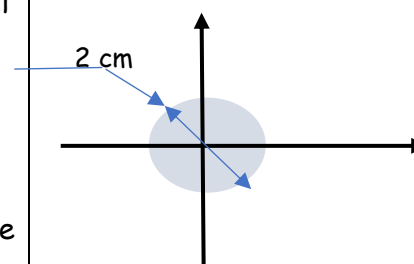
$$I_{3xx} = \frac{\pi r^4}{4} = \frac{\pi \times 1^4}{4} = 0.78 \text{ cm}^4$$

$$I_{3yy} = \frac{\pi r^4}{4} = \frac{\pi \times 1^4}{4} = 0.78 \text{ cm}^4$$

Since the circular area is removed, its MI is subtracted from the other two.

$$I_{xx} = I_{1xx} + I_{2xx} - I_{3xx} = 26.83 \text{ cm}^4$$

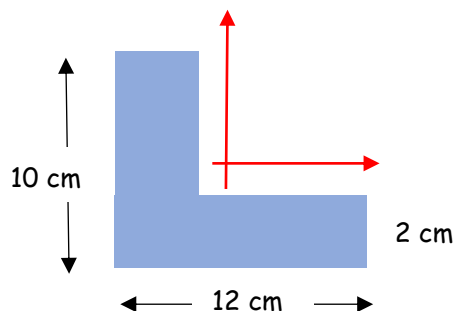
$$I_{yy} = I_{1yy} + I_{2yy} - I_{3yy} = 16.17 \text{ cm}^4$$



#Exercise 19

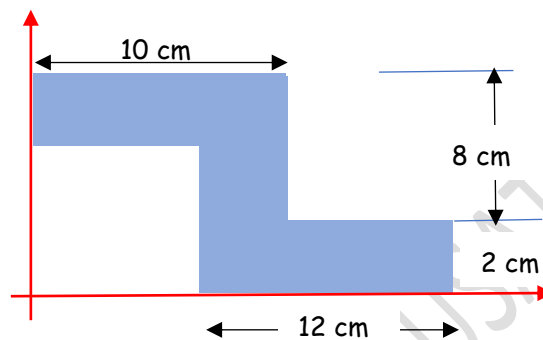
19.1

Calculate the moment of inertia of the given shape with respect to its **centroidal** x-y axes.



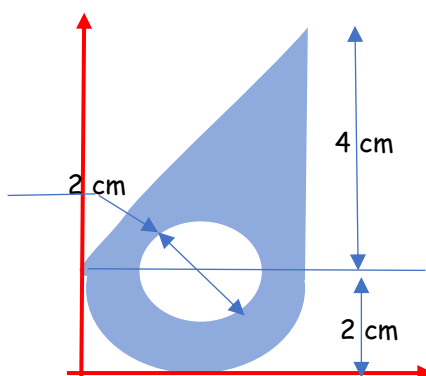
19.2

Calculate the moment of inertia of the given shape with respect to the given x-y axes.



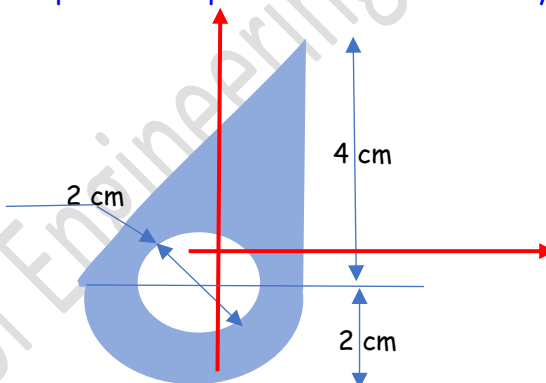
19.3

Calculate the moment of inertia of the given shape with respect to its x-y axes.



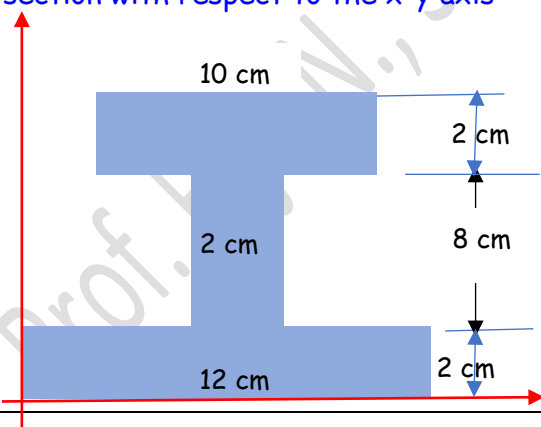
19.4

Calculate the moment of inertia of the given shape with respect to its **centroidal** x-y axes.



19.5

Calculate the moment of inertia of the I section with respect to the x-y axis



19.6

Calculate the moment of inertia of the I-section given in #Exercise 19.5 with respect to its centroidal x-y axes.

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